6.1

a.) Why should the data be partitioned into training and holdout sets? What will the training set be used for? What will the holdout set be used for?

Answer: In order to accurately assess model performance, the data has to be partitioned into training and holdout sets. The training set is employed to fit the model and alter the parameters, whereas the holdout set is used solely for final validation. This means that the performance metrics indicates the model's ability to generalize to unseen data,rather than simply memorizing the training examples.

b.) Fit a multiple linear regression model to the median house price (MEDV) as a function of CRIM, CHAS, and RM. Write the equation for predicting the median house price from the predictors in the model.

In [1]:
```
import matplotlib.pyplot as plt # for plotting
import mlba
import numpy as np # for numerical data handling
import pandas as pd # for tabular data handling
import statsmodels.formula.api as sm # for statistical modeling
```

```
no display found. Using non-interactive Agg backend
```

In [2]:
```
#VB67
# Loading Boston Housing data from mlba
housing_df = mlba.load_data('BostonHousing.csv')
```

In [3]:
```
#VB67
#displaying first five rows
housing_df.head()
```

Out[3]:

	CRIM	ZN	INDUS	CHAS	NOX	RM	AGE	DIS	RAD	TAX	PTRATIO	LS
0	0.00632	18.0	2.31	0	0.538	6.575	65.2	4.0900	1	296	15.3	
1	0.02731	0.0	7.07	0	0.469	6.421	78.9	4.9671	2	242	17.8	
2	0.02729	0.0	7.07	0	0.469	7.185	61.1	4.9671	2	242	17.8	
3	0.03237	0.0	2.18	0	0.458	6.998	45.8	6.0622	3	222	18.7	
4	0.06905	0.0	2.18	0	0.458	7.147	54.2	6.0622	3	222	18.7	

In [4]:

```
#VB67

# housing_df["MEDV"]

# Selecting predictors and target
X = housing_df[['CRIM', 'CHAS', 'RM']]
y = housing_df['MEDV']

# Add constant (intercept)
# X = sm.add_constant(X)

# Fit regression model
model = sm.ols("MEDV ~ CRIM + CHAS + RM", data=housing_df).fit()

# Printing regression summary
print(model.summary())

# Printing regression equation
print("\nRegression Equation:")
print(f"MEDV = {model.params['Intercept']:.2f} "
      f"+ ({model.params['CRIM']:.2f} * CRIM) "
      f"+ ({model.params['CHAS']:.2f} * CHAS) "
      f"+ ({model.params['RM']:.2f} * RM)")
```

```
                            OLS Regression Results
==============================================================================
==
Dep. Variable:                   MEDV   R-squared:                       0.5
53
Model:                            OLS   Adj. R-squared:                  0.5
50
Method:                 Least Squares   F-statistic:                     20
6.7
Date:                Wed, 24 Sep 2025   Prob (F-statistic):           2.73e-
87
Time:                        18:27:23   Log-Likelihood:                -163
6.7
No. Observations:                 506   AIC:                             328
1.
Df Residuals:                     502   BIC:                             329
8.
Df Model:                           3
Covariance Type:            nonrobust
==============================================================================
==
                 coef    std err          t      P>|t|      [0.025      0.97
5]
------------------------------------------------------------------------------
--
Intercept    -28.8107      2.563    -11.240      0.000     -33.847     -23.7
75
CRIM          -0.2607      0.033     -7.964      0.000      -0.325      -0.1
96
CHAS           3.7630      1.086      3.464      0.001       1.629       5.8
97
RM             8.2782      0.402     20.602      0.000       7.489       9.0
68
==============================================================================
==
Omnibus:                      157.437   Durbin-Watson:                   0.8
75
Prob(Omnibus):                  0.000   Jarque-Bera (JB):              944.6
66
Skew:                           1.216   Prob(JB):                     7.38e-2
06
Kurtosis:                       9.236   Cond. No.                         9
2.5
==============================================================================
==

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is corre
ctly specified.

Regression Equation:
MEDV = -28.81 + (-0.26 * CRIM) + (3.76 * CHAS) + (8.28 * RM)
```

C.) Using the estimated regression model, what median house price is predicted for a tract in the Boston area that does not bound the Charles River, has a crime rate of 0.1, and where the average number of rooms per house is 6?

In [7]:
```
#VB67
# Predicting the target value using the trained model for a new sample with
q_df =  pd.DataFrame({
    "CRIM": [0.1],
    "CHAS": [0],
    "RM": [6]
})
pred_output = model.predict(q_df)
```

In [6]:
```
#VB67
# Displaying the predicted MEDV in $1000s and actual dollar value for the ne

print(f"Predicted MEDV:{pred_output[0]:.1f}(in $1000s)")
print(f"Median value of owner-occupied home is ${pred_output[0]*1000:,.0f}")
```

```
Predicted MEDV:20.8(in $1000s)
Median value of owner-occupied home is $20,832
```

In [ ]:

In [ ]:

In [ ]: